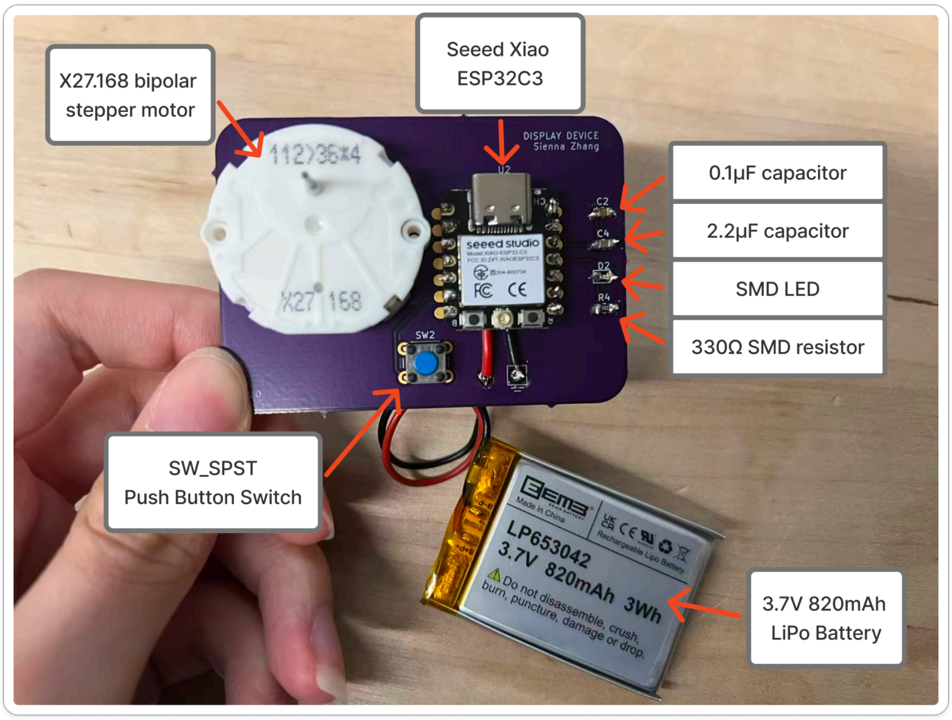
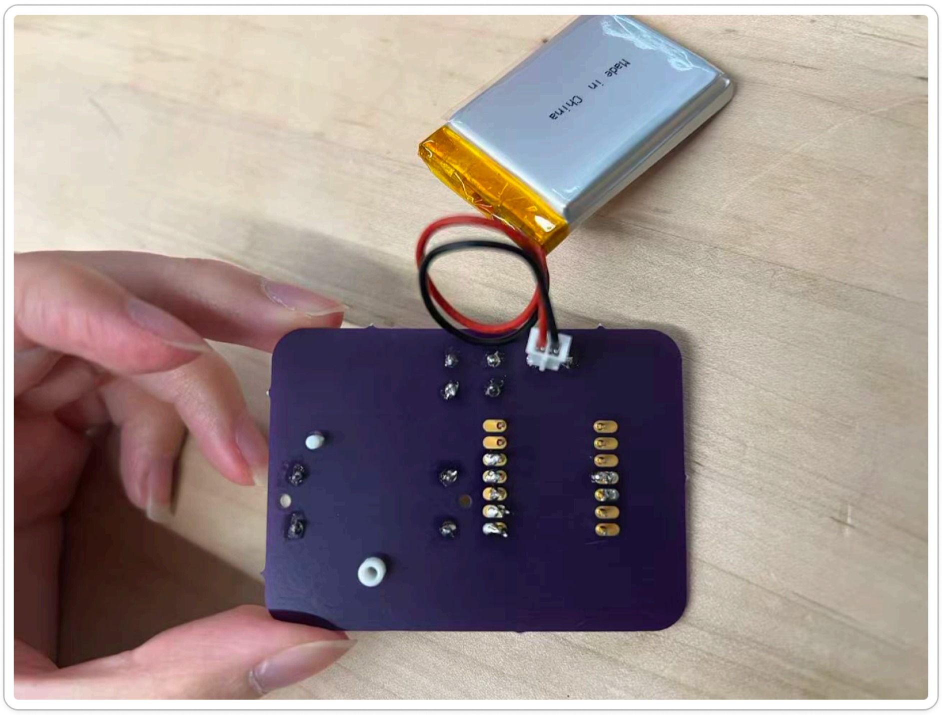
Assembled PCBA





I conducted functional testing of the power system and overall device operation. During testing, I found that the power connection was unstable. The battery footprint spacing on the PCB is too wide, which makes the soldered connection unreliable and requires additional copper wires to bridge the gap. For future revisions, the battery pads should be repositioned directly underneath the ESP32 module so the battery can be soldered securely without extra wiring.

I also identified a design limitation in the power control. The current push button only switches modes and does not function as a power switch. As a result, there is no dedicated hardware power control on the board. Additionally, in the previous PCB revision, the connection between the button and the ESP32 was missing and required a fly wire to complete the circuit.

For the next iteration, improvements should include optimizing the battery pad placement, integrating a proper power switch into the circuit design, and ensuring all button-to-ESP connections are correctly routed in the PCB layout to avoid manual rework.